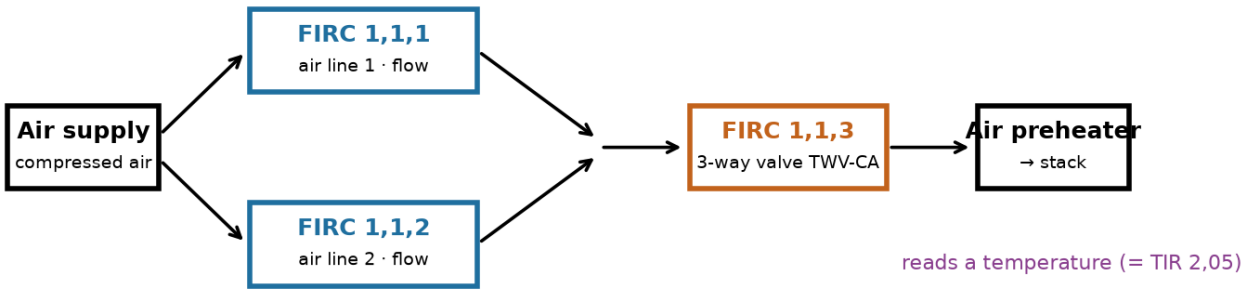
How the three FIRC signals move together

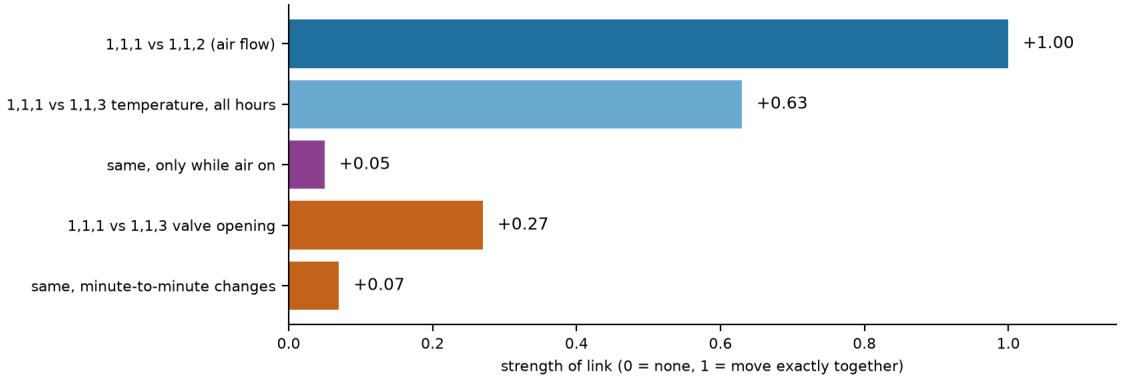
SOFC air supply · FIRC 1,1,1 / 1,1,2 / 1,1,3 · data: 39 logged days, 1-minute averages · 2 Nov 2026

1. The answer

FIRC 1,1,1 and 1,1,2 are twins: they move in perfect step (correlation 1.00). **FIRC 1,1,3 barely follows them**, because it is not an air-flow reading at all. It is a temperature controller for a 3-way valve.

Why it matters: one of the twin air signals is enough for stage rules and synthetic data, and FIRC 1,1,3 must not be treated as a third air flow.



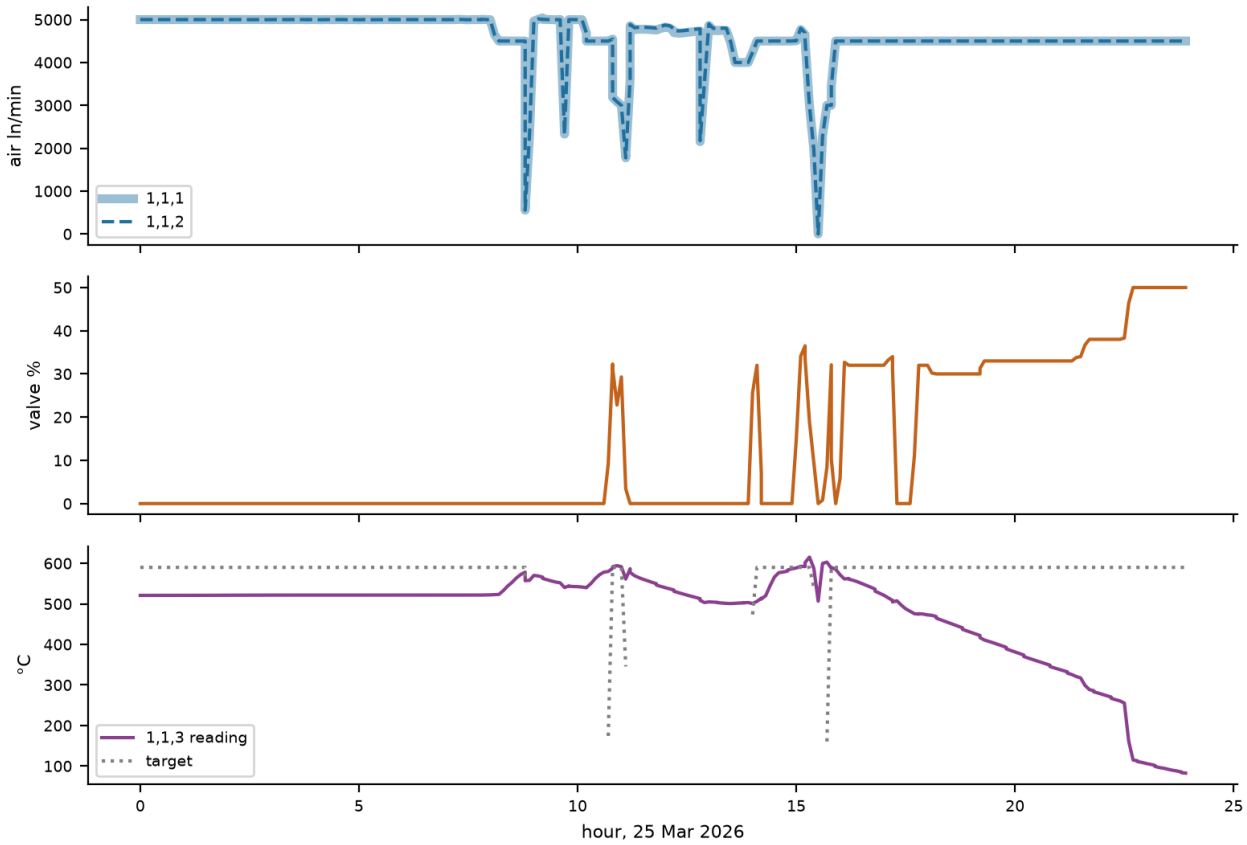


Pair	Correlation	Meaning
1,1,1 - 1,1,2	1.00	same signal; ratio 1.000, typical gap 0
1,1,1 - 1,1,3 reading	0.63 -> 0.05	only plant on/off in common
1,1,1 - 1,1,3 valve	0.27	weak; step changes 0.07
1,1,3 valve - its temperature gap	-0.66	its real driver

If you remember one thing: FIRC 1,1,1 = FIRC 1,1,2. FIRC 1,1,3 follows temperature, not air.

2. FIRC 1,1,1 and 1,1,2 are one signal twice

The plant splits the air into two equal lines with the same target, so both readings rise, hold and fall together.



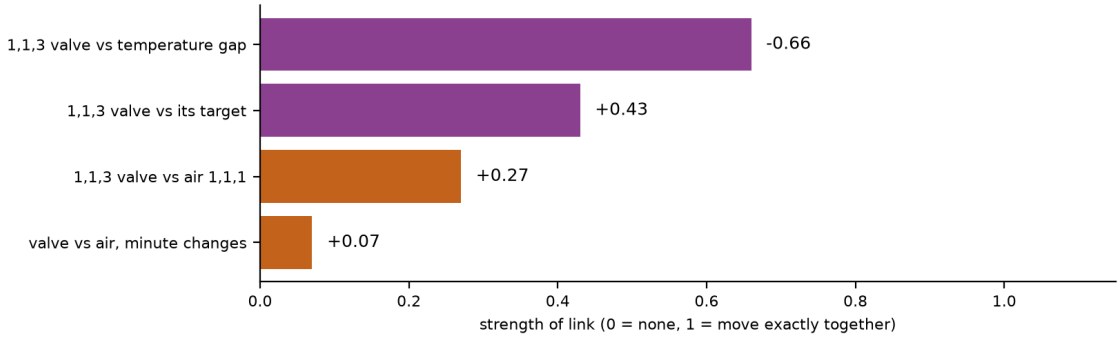
Check (44,341 minutes with air on)	Result
Correlation	1.000
Ratio 1,1,2 / 1,1,1, middle half	0.9999 - 1.0001
Typical gap	0 ln/min (spread 4.8)
Rank correlation, plant-off hours included	0.83 (noise near zero only)

Use one of them (or their sum). If they ever split apart, raise an alarm: a blocked line or failed controller.

If you remember one thing: the two air lines carry the same air, so their readings are the same number.

3. FIRC 1,1,3 follows temperature, not air

Its reading equals TIR 2,05 (a temperature) exactly. Its output opens a 3-way valve when that temperature sits below its 570 / 590 °C target. Logged only 17-30 Mar 2026 (16,193 minutes).



The 0.63 over all hours comes from both being zero when the plant is off. With air on it drops to 0.05. Shifting by up to 30 minutes either way gives only 0.27 - 0.31.

Air level (ln/min)	Minutes	Valve open	Avg opening
0 (off)	3,056	0 %	0 %
~3000	5,742	3 %	1 %
~4000	1,352	69 %	28 %
~5000	6,004	28 %	10 %

Those air levels are used while heating and adjusting, when the gas is still below target; when the valve is open the temperature is typically 317 °C under target. Same plant state seen twice, not air driving the valve.

Signals: ct_FIRC_111/112_PV_ProcessValue, ct_FIRC_113_PV_ProcessValue (= TIR_2_05), ct_FIRC_113_OUT_OutputValue, ct_FIRC_113_SP_Analog. PLC_ctrl_ct_FIRC_113 is noise.

If you remember one thing: FIRC 1,1,3 is a temperature-driven valve. Don't use it as a third air flow.